

A Method for Passive Identification of Spacecraft 1

Using Doppler-Based Trajectory Matching 2

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Abstract 6

This paper presents a passive method for identifying unknown spacecraft using 7
the Doppler signature of a single satellite pass, requiring neither training data 8
nor the precise knowledge of the transmitter carrier frequency. Doppler shifts 9
are extracted from SDR IQ samples via a phase-locked loop and averaged into 10
a one second time series. Candidates from an open Two-Line Element catalog 11
are processed by a seven-stage filtering algorithm before predicted Doppler tra- 12
jectories are generated using a computationally efficient combination of SGP4 13
propagation and F&G (Lagrange coefficient) functions. Observed and predicted 14
series are independently mean-centered to cancel unknown carrier offsets and 15
SDR oscillator instability, and candidates are ranked by root-mean-square error. 16
Using consumer-grade SDR receivers, the method achieved 28 successful single 17
pass identifications across the VHF, UHF, L-, and S-bands, with the correct 18
satellite ranking first in every case and an RMSE typically several times lower 19
than the nearest competing candidate. The implementation is released as open- 20
source software. These results show that accurate, low-cost passive spacecraft 21
identification is achievable without specialized infrastructure or prior calibration. 22

Keywords: Passive satellite identification, Doppler frequency shift, single pass 23
identification, Software-defined radio, Space situational awareness, Non-cooperative 24
identification, Two-Line Element set, SGP4, F&G functions 25

Declarations. 26

27 **Availability of data and materials.** The source code of the Dopp-
28 lerguesser system developed during this study is publicly available at
29 <https://github.com/BUSH222/dopplerguesser> (Vtorov 2026). The software is released
30 under the MIT License. The raw IQ recordings, frequency time series, and screen
31 capture files supporting the conclusions of this article are available from the
32 corresponding author on reasonable request.

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35 **Authors' contributions.** Fedor Ivanovich Vtorov devised the method, imple-
36 mented the software, conducted all observations, performed the analysis, and wrote
37 the manuscript.

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39 **References**

40 Vtorov FI (2026) Dopplerguesser. <https://github.com/BUSH222/dopplerguesser>,
41 source code of the Dopplerguesser system developed during this study. Accessed:
42 2026-07-01